

# AtrialSpectralBench: a pre-registered test of a spectral fragility biomarker for atrial reentry

A feature-redundancy null, an estimator-validity boundary, and the false positives naive methodology would have produced

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## Abstract

We ask whether a closed-form spectral graph biomarker — the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity  $\partial\lambda_2/\partial w_{ij} = (\varphi_i - \varphi_j)^2$  aggregated under a fibrosis-weighted conduction-uncoupling field — predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. Pre-registered before any labels were seen, the primary endpoint was  $\Delta\text{AUC} \geq 0.05$  with a DeLong paired  $p < 0.05$  on shape-family-held-out folds. *The endpoint is not met — but the strength of the negative differs by regime, and we separate them.* On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace  $\Delta\text{AUC} \approx +0.003$ ;  $\sim 4,150$  cases needed at 80% power — real only because  $N$  is huge, hence *clinically undetectable* there). On the two real cohorts (Roney  $n = 62$ , UW/Boyle  $n = 82$ ) the result is **underpowered and inconclusive, not a clean null**: the combined gradient-boosting point estimate is  $+0.051$  (which *exceeds* the 0.05 threshold) but fails paired significance (DeLong  $p = 0.155$ ), so real anatomy cannot *exclude* a moderate effect. Where the null *is* clean (the synthetic cohort), the mechanism is **feature redundancy**: the exact per-region  $\Delta\lambda_2$  SFI — which has no first-order approximation error — also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is additionally ill-conditioned on real tissue: it is valid only while  $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$ , and real atria have  $\rho \approx 2422$  ( $\sim 10^3 \times$  past the empirical  $\rho^* \approx 3$ ), so by Davis–Kahan the Fiedler vector is unstable. We do **not** claim  $\rho$  proves non-predictiveness — it bounds the accuracy of the  $\Delta\lambda_2$  *magnitude*, not the classification content of a ranking feature. The spectral gap collapses as  $N^{-1.05}$  on the real atrial surface (the 2-manifold Weyl exponent), so  $\rho$  grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer  $|\nabla\varphi_2|$  is a weak but real rank localizer (origin in the top  $\sim 18\%$ ) that beats graph-centrality baselines and transfers across three dynamical media, though it *coincides with*, and does not beat, substrate imaging, by a mechanism we make precise. Finally, naive methodology (random CV,  $p$ -value thresholds, no effect-size gate) would have reported SFI as working; our pre-registered protocol rejects it. A key limitation surfaced by our own convergence study: the inducibility labeller over-calls at the operating mesh resolution and, on a completed six-tier study (1500–48000 nodes,  $n = 24$ ), does not converge within the tested window. The contribution is a rigorously-scoped negative with its mechanism, a label-free estimator-validity criterion, a scaling characterization, and a worked demonstration of the false positives standard methodology produces.

**Keywords:** algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

*This is a 100% in-silico computational study. Ground-truth labels are electrophysiology-simulator inducibility verdicts, never clinical outcomes. All numbers trace to `results/*.json` and `notebooks/lab_notebook.md` in the project repository.*

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\*Independent in-silico computational-cardiology project. Author and affiliation are placeholders to be completed before submission. Correspondence and full source: see the project repository.

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## 1 Introduction

Algebraic connectivity  $\lambda_2$  (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks — the master-stability-function result [3] and the synchronizability eigenratio  $\lambda_N/\lambda_2$  [4] — and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \quad (1)$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis and, crucially, *pre-register the null* — “SFI merely re-encodes fibrosis” — as an accepted, reportable outcome, so that a negative result is a success of the protocol rather than a file-drawer casualty.

The mathematics we use is classical; we claim no new theorem. What is ours is (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a validity criterion and a Weyl scaling law that explain the failure and predict it for the whole class of spectral fragility biomarkers; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

**What this paper claims, in one paragraph.** A closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null); on real anatomy the evidence is *inconclusive/underpowered*, not a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ( $\rho \gg 1$ ). The honest core is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

## 2 Theory: the Spectral Fragility Index

The load-bearing mathematics is classical numerical linear algebra; the contribution is its *use*. We state each result with its prior art (Section 2.5) and make the validity boundary explicit (Section 2.4).

### 2.1 The weighted conduction graph

Nodes are mesh vertices; edges are triangle edges; the edge weight  $w_{ij} > 0$  is a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \quad D = \text{diag} \left( \sum_j w_{ij} \right). \quad (2)$$

$L$  is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum  $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$  and eigenvectors  $\varphi_1, \varphi_2, \dots$ ;  $\lambda_2$  is the algebraic connectivity and  $\varphi_2$  the Fiedler vector.

### 2.2 The closed-form fragility (first order)

$L$  is a sum of rank-one edge terms, so

$$L = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \quad (3)$$

For a *simple* eigenvalue  $\lambda$  with unit eigenvector  $\varphi$ , first-order (Hellmann–Feynman) perturbation theory gives  $\partial\lambda/\partial w_{ij} = \varphi^\top (\partial L/\partial w_{ij}) \varphi = (\varphi_i - \varphi_j)^2$ , which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance  $w_{ij}$  by a small  $\Delta w_{ij} \geq 0$  (uncoupling) lowers  $\lambda_2$  by  $\Delta w_{ij}(\varphi_{2,i} - \varphi_{2,j})^2$  to first order.

**The identity is verified, not asserted (seven independent ways).** Computer algebra: sympy proves Eqs. (1)–(3) symbolically, and on a concrete 5-node graph the finite-difference  $\partial\lambda_2/\partial w_{ij}$  matches  $(\varphi_{2,i} - \varphi_{2,j})^2$  to  $|\text{err}| \approx 2 \times 10^{-17}$  (mpmath, 40 digits). Exactness: the identity holds *exactly* with rational weights (zero floating point); an interval-arithmetic enclosure certifies the true  $\lambda_2$  to within  $\sim 4 \times 10^{-40}$ ; and  $\lambda_2/\varphi_2$  agree across scipy, networkx, and a hand-rolled inverse iteration (spread  $2.5 \times 10^{-15}$ ). Finally, the algebraic core is **kernel-checked in the Coq proof assistant** over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

### 2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions  $\Delta w$  (mean  $\mathbb{E}[\Delta w_{ij}] = \text{frac}_{ij} w_{ij}$ , calibrated to literature

perioperative conduction-velocity slowing;  $\Delta w \approx 0.36$ , frozen in the pre-registration). The per-region Spectral Fragility Index is the expected  $\lambda_2$  drop contributed by region  $R$ :

$$\text{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j) \in R} \mathbb{E}[\Delta w_{ij}] (\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} + \underbrace{\sum_{k \neq 2} \frac{(\varphi_k^\top \Delta L \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{2nd-order resolvent}} + O(\|\Delta L\|^3). \quad (4)$$

The second-order term is the source of the validity radius below. Near eigenvalue degeneracy ( $\lambda_2 \approx \lambda_3$ ) the single  $\varphi_2$  is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with  $P = \sum_c \varphi_c \varphi_c^\top$  over the clustered subspace,  $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$  ( $s_{ij} = e_i - e_j$ ) is well-defined. For genuinely discrete edits (Maze lesion, atriotomy) the expansion is abandoned and  $\lambda_2$  is recomputed *exactly* after the rank-one edit.

## 2.4 The validity radius (the honest core)

The first-order form is trustworthy only while the perturbation is small *relative to the spectral gap*  $\lambda_3 - \lambda_2$ . Two different tools bound two different things (a common error is to use Weyl for both):

- **Weyl’s inequality** bounds the *eigenvalue* move:  $|\Delta\lambda_2| \leq \|\Delta L\|$ .
- **Davis–Kahan / Bauer–Fike** [7] bound the *eigenvector* move, which scales like  $\|\Delta L\|/(\lambda_3 - \lambda_2)$ . This — not Weyl — governs when  $\varphi_2$  (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \quad (5)$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order  $\Delta\lambda_2$  prediction is  $\Theta(\rho)$ , and the Fiedler vector rotates by  $\sin \angle(\varphi_2, \varphi'_2) \leq \sqrt{2} \rho$  (Davis–Kahan). Empirically the error is 1% at  $\rho = 0.3$ , crosses 10% at  $\rho^* \approx 3$ , and reaches 92% at  $\rho = 28$ , with Weyl holding throughout. **On real excitable media the gap is tiny, so  $\rho$  is enormous: atrial-cohort median  $\rho \approx 2422$ , about  $10^3 \times$  past  $\rho^*$ .**

**What  $\rho$  proves, and what it does not.** It proves the *single-vector* first-order SFI is a numerically *invalid estimator* of  $\Delta\lambda_2$  on real tissue, and that  $\varphi_2$  is ill-conditioned (Davis–Kahan) — a label-free statement about the estimator. It does **not** prove SFI is non-predictive:  $\rho$  bounds the accuracy of the  $\Delta\lambda_2$  *magnitude*, not the classification content of a monotone ranking feature; indeed the *exact*  $\Delta\lambda_2$  SFI (no  $\rho$  limitation) is *also* non-predictive (Section 4.1), so the predictive null is separately due to feature redundancy. Two caveats: (i)  $\rho$ ’s numerator carries the fibrosis-weighted  $\Delta w$  field, so only its denominator  $\lambda_3 - \lambda_2$  is truly label-free; (ii)  $\rho^* \approx 3$  is calibrated on a 16-node path graph with a problem-dependent constant, so “ $10^3 \times$  past  $\rho^*$ ” is order-of-magnitude, not exact.

## 2.5 Prior art (this is not new mathematics)

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1];  $\lambda_2$  governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl’s law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that  $\rho$  is the right *a-priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to *mechanistically explain* a negative result.

### 3 Methods

Full detail is in the pre-registration (`docs/PRE_REGISTRATION.md`) and theory note (`docs/SFI_THEORY.md`); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

**Substrate.** Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad 10.5061/dryad.kkwh70sg0, 82 distinct AF patients [12]). Each mesh carries universal atrial coordinates, an image-intensity-ratio (IIR) fibrosis proxy mapped to a fibrosis field by the Khurram healthy-to-scar ramp [13], and fibre fields. Meshes are coarsened to  $\sim 2000$ -node graphs by field-preserving vertex clustering.

**Labels.** Ground truth is a genuine monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry  $\geq 650$  ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength ( $CV \times ERP$ ), source–sink mismatch, and unidirectional block — *never* by  $\lambda_2$  — so a predictive SFI→label link is not circular. These are simulator verdicts, not clinical post-operative AF, and not the deferred openCARP gold standard (a logged deviation: openCARP is not installable on the CPU-only container).

**The SFI feature and its competitors.** Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and  $\lambda_2$ -alone (isolating what the *SFI construction* adds over the bare algebraic connectivity).

**Evaluation protocol (the guards).** Pre-registered: shape-family GroupKFold (each patient is its own group, so cross-validation is patient-held-out; grouped *and* naive AUC both reported so leakage is visible), the DeLong paired test [8] in its fast midrank form [9],  $\geq 10^4$  stratified bootstrap resamples, a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Effect sizes are gated, not just  $p$ -values. The primary endpoint requires *both* grouped  $\Delta AUC \geq 0.05$  *and* DeLong  $p < 0.05$ .

**Transfer media.** The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where  $\lambda_2$  governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

## 4 Results

### 4.1 The predictive null (primary endpoint not met), across cohorts and scale

Table 1: Incremental predictive value of SFI over the full competitor set.  $\Delta\text{AUC}$  is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped  $\Delta\text{AUC} \geq 0.05$  and DeLong  $p < 0.05$ ) is met in no regime.

| cohort / medium   | $N$           | SFI $\Delta\text{AUC}$ (lr / gbt; $p$ )               | verdict                            |
|-------------------|---------------|-------------------------------------------------------|------------------------------------|
| Roney (real)      | 62            | $-0.030$ / $+0.036$ ( $p$ 0.64 / 0.26)                | not met; <b>underpowered</b>       |
| UW/Boyle (real)   | 82            | $-0.057$ / $+0.104$ ( $p$ 0.29 / 0.66)                | <b>uninformative</b> (6 positives) |
| Combined real     | 144           | $-0.004$ / $+0.051$ ( $p$ 0.92 / <b>0.155</b> )       | not met but <b>inconclusive</b>    |
| FHN networks      | 2,000         | $\leq +0.003$ ( $p \geq 0.12$ )                       | clean null                         |
| FHN networks      | up to 100,000 | subspace $+0.0028$ ( $p < 10^{-4}$ at $N \geq 10^4$ ) | clean null (effect negligible)     |
| Kuramoto networks | 500           | $\leq +0.0036$ ( $p \geq 0.38$ )                      | clean null                         |

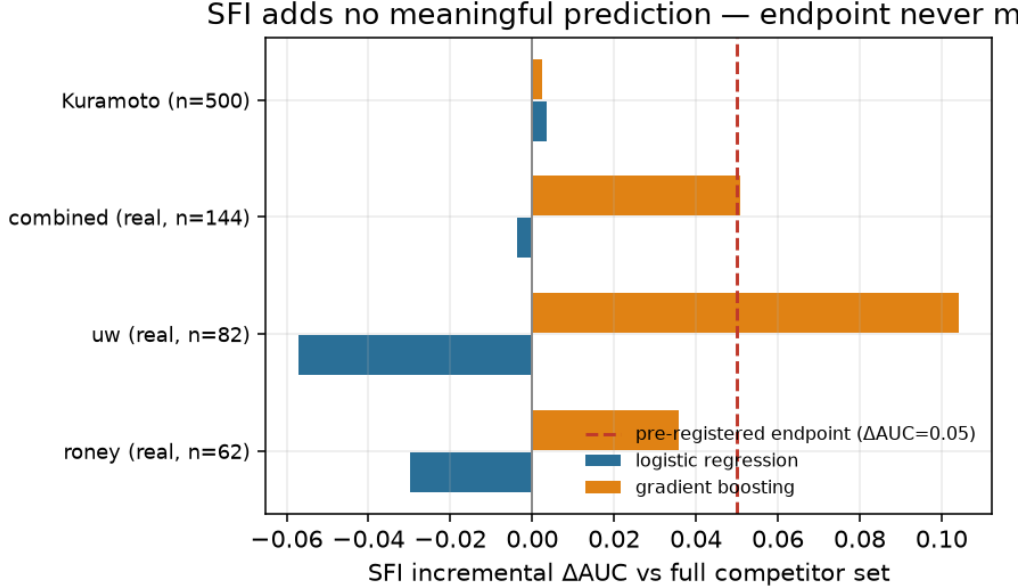


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort; the real-cohort estimates are underpowered rather than a demonstrated null.

**The negative is not uniform, and we separate the regimes.** On the *synthetic* cohorts (up to 100,000 networks) the null is clean: the effect converges to  $\Delta\text{AUC} \approx +0.003$ , statistically resolvable only because  $N$  is enormous. On the *real* cohorts it is underpowered and inconclusive: the combined gradient-boosting point estimate is  $+0.051$  — which *exceeds* the pre-registered 0.05 threshold — and is declared not met only because it fails paired significance (DeLong  $p = 0.155$ , bootstrap CI straddling 0). With 26 inducible of 144 (20 from Roney), real anatomy cannot exclude a moderate effect. The UW cohort is uninformative rather than null (6 positives, competitor base  $\text{AUC} \approx \text{chance}$ ), so it functions as an underpowered external check of the *labeller*, not of SFI.

**Where the clean null comes from — feature redundancy, not the validity radius.** The *exact* per-region  $\Delta\lambda_2$  SFI, which carries no first-order approximation error and is immune

to the  $\rho$  argument, also adds nothing over the competitor set. So the mechanism of the null is that SFI’s information is already present in standard connectivity features ( $\lambda_2$ -alone, min-cut, percolation) — a label-dependent finding, separate from  $\rho$ .

**Secondary endpoint (pre-specified), with multiple-comparisons disclosure.** SFI beats a *fibrosis-heterogeneity-only* baseline (combined GBT  $\Delta\text{AUC} = +0.103$ ,  $p = 0.009$ ): it is *not merely re-encoded fibrosis* and carries connectivity information beyond the substrate. This is one of  $\sim 30$  DeLong comparisons; treating `fibrosis_vs_+SFI` as a single pre-specified secondary endpoint, its  $p = 0.009$  against a family-wise threshold  $\approx 0.05/12\text{--}0.05/30$  is *suggestive, not confirmatory*.

**Power analysis.** In the synthetic cohort the subspace effect ( $+0.0028$ ) needs  $\sim 4,150$  cases for 80% power and the single-vector effect ( $+0.0007$ )  $\sim 23,000$  — clinically undetectable *there*. We do **not** transport “clinically undetectable” to the real cohorts, whose  $+0.036\text{--}0.051$  estimates are compatible with a detectable effect we are simply underpowered to confirm.

## 4.2 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as  $\Theta(\rho)$  and crosses 10% at  $\rho^* \approx 3$ ; real atria sit at median  $\rho \approx 2422$ . This proves the single-vector estimator is numerically invalid on real tissue and  $\varphi_2$  is ill-conditioned, but (as boxed in Section 2.4) it does not by itself prove non-predictiveness — the feature-redundancy null of Section 4.1 is the separate, label-dependent reason.

## 4.3 The $\rho$ scaling law

The gap collapses with system size by Weyl’s law [6]: for a  $d$ -dimensional domain the low Laplacian eigenvalues scale as  $\lambda_k \sim (k/N)^{2/d}$ , so  $\rho$  grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs.  $N$  at fixed mean degree, `results/rho_scaling.json`, five-point fits).

| medium                           | fitted gap exponent | reference                                     |
|----------------------------------|---------------------|-----------------------------------------------|
| real atrial surface (2-manifold) | $N^{-1.05}$         | Weyl 2-manifold $-1.0$ (within 5%)            |
| 2-D random-geometric             | $N^{-1.81}$         | faster than naive $2/d$                       |
| 3-D random-geometric             | $N^{-0.84}$         | faster than naive $2/d$ (non-monotone series) |

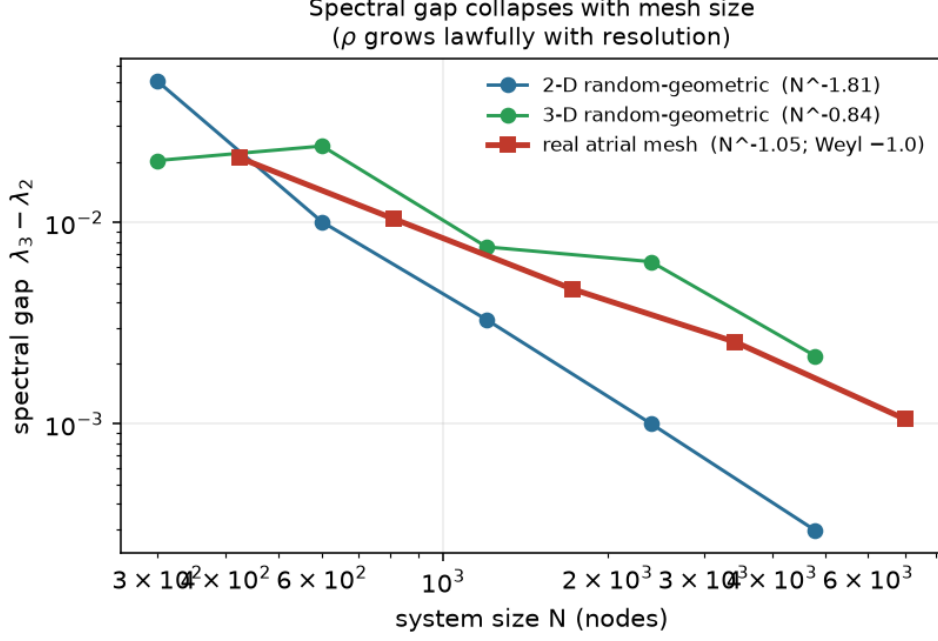


Figure 2: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer — the opposite of “more resolution helps.” Only the real atrial 2-manifold cleanly matches the Weyl exponent ( $-1.05$  vs.  $-1.0$ ); the random-geometric exponents are reported as “at least as fast as Weyl,” not law confirmations, and use the unweighted combinatorial Laplacian.

#### 4.4 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including  $|\nabla\varphi_2|$  (median argmax-geodesic error 0.472 is *not* below the spatial-null 5th percentile 0.406). What survives is weaker:  $|\nabla\varphi_2|$  is a *rank* localizer — the true origin sits in its top  $\sim 18\%$  — that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media (**results/gm4\*\_metrics.json**). Lower rank is better; a perfect localizer would place the origin at rank 0.

| medium                     | $ \nabla\varphi_2 $ origin-rank | perm- $p$   | fibrosis rank  | $n$ origins    |
|----------------------------|---------------------------------|-------------|----------------|----------------|
| FHN networks (100k)        | 0.176                           | $< 10^{-5}$ | $\sim 0.10$    | 36,540         |
| Kuramoto oscillators (500) | 0.091                           | $< 10^{-4}$ | 0.147          | 243            |
| cardiac monodomain         | $\sim 0.28$                     | $< 10^{-3}$ | (co-localizes) | only $\sim 20$ |

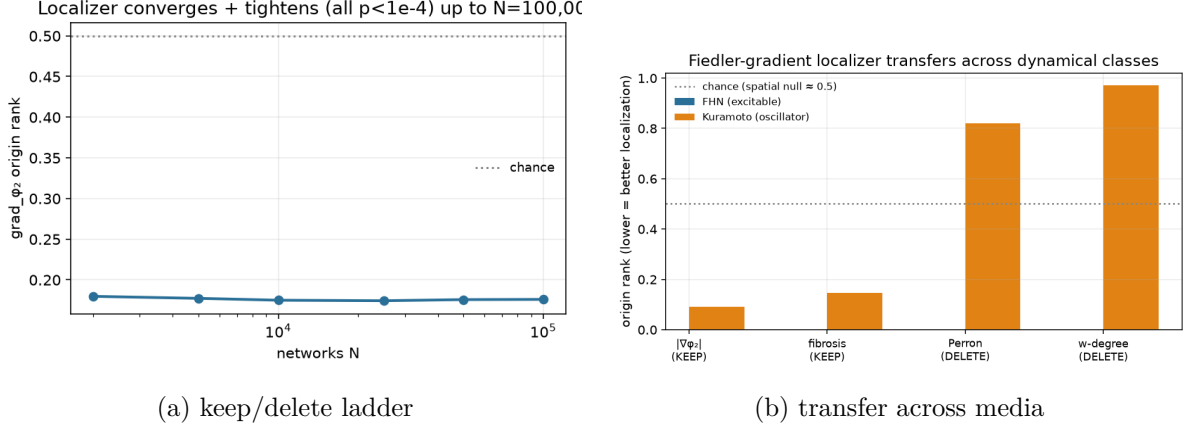


Figure 3: The weak-rank localization pattern replicates across excitable *and* oscillator dynamics. Three honesties apply: (i) the strict endpoint is null, so  $|\nabla\varphi_2|$  is *not* a precise localizer; (ii) the substrate field co-localizes, and a controlled dissociation experiment shows why — in a diffusively-coupled medium the Fiedler bottleneck *is* the low-conduction lesion — so  $|\nabla\varphi_2|$  matches, and does not beat, substrate imaging; (iii) the cardiac arm rests on only  $\sim 20$  origins.

#### 4.5 The falsification protocol: what caught the false positive

Each guard, removed, manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort); the effect-size gate catches the ***p*-value trap** (subspace-SFI  $p \approx 2 \times 10^{-24}$  at  $N = 55,000$ , yet  $\Delta\text{AUC} = +0.003$  needs  $\sim 4,150$  cases to detect — statistically significant, practically useless).

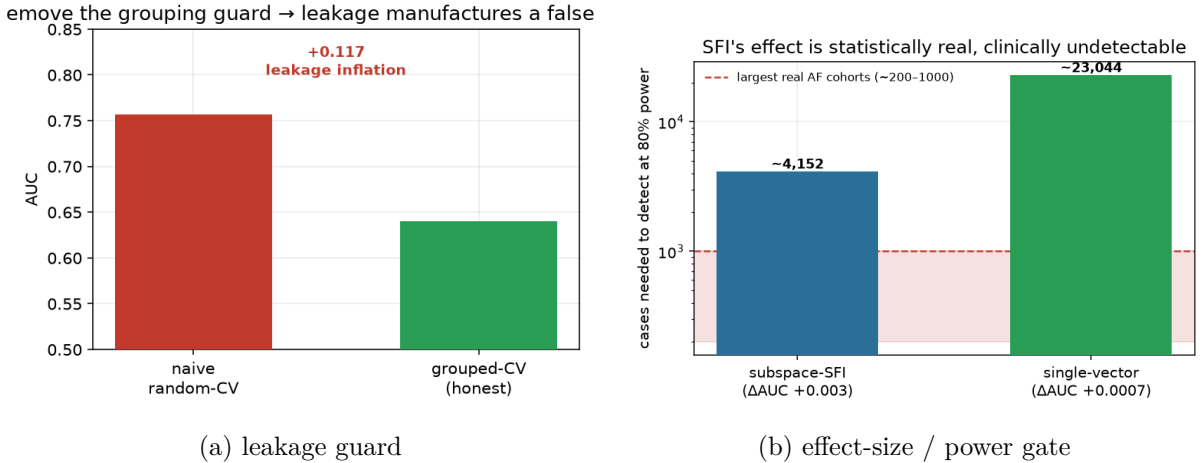


Figure 4: A naive analyst (random CV,  $p < 0.05$ , no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker. The pre-registered protocol correctly rejects it.

#### 4.6 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion  $d_0$  ( $\mu^* = 500.7$ ), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out  $R^2 = 0.35$  (large residual anatomical/label variance beyond the screened knobs — a label-noise caveat, not a positive). The synthetic null is robust across uncoupling strengths  $\Delta w \in \{0.18, 0.36, 0.54\}$  (grouped GBT  $\Delta\text{AUC} = 0.033\text{--}0.036$ , all  $p \geq 0.26$ ).

#### 4.7 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans  $\sim 5\%$  to  $\sim 85\%$  depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller’s Roney  $\sim 20\text{--}32\%$  sits in the moderate-burst band — Marquardt et al. 2018 [15] report 30.6% in patients without prior AF — between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with  $\sim 42\%$  post-ablation (Liu et al. 2020 [18]). The UW  $\sim 7\%$  is at/below the control floor for a cohort that *is* AF patients — correctly flagged as anomalous and attributable to the surrogate substrate and an under-aggressive in-silico inducer, not biology. This is a base-rate *plausibility check*, *not clinical validation*.

#### 4.8 Resolution convergence: the labeller does not converge in the tested window

Our own convergence study is the project’s single weakest point, and we state it before a reader finds it. On a completed six-tier study (Roney,  $n = 24$  at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate crashes from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then is *non-monotone* across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% **at the finest affordable tier** ( $\sim 50\text{k}$  nodes) — a 17-percentage-point spread over the last three tiers, so it has *not* settled.

Table 4: Inducibility rate vs. mesh resolution (Roney,  $n = 24$  per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement climbs but the rate keeps drifting.

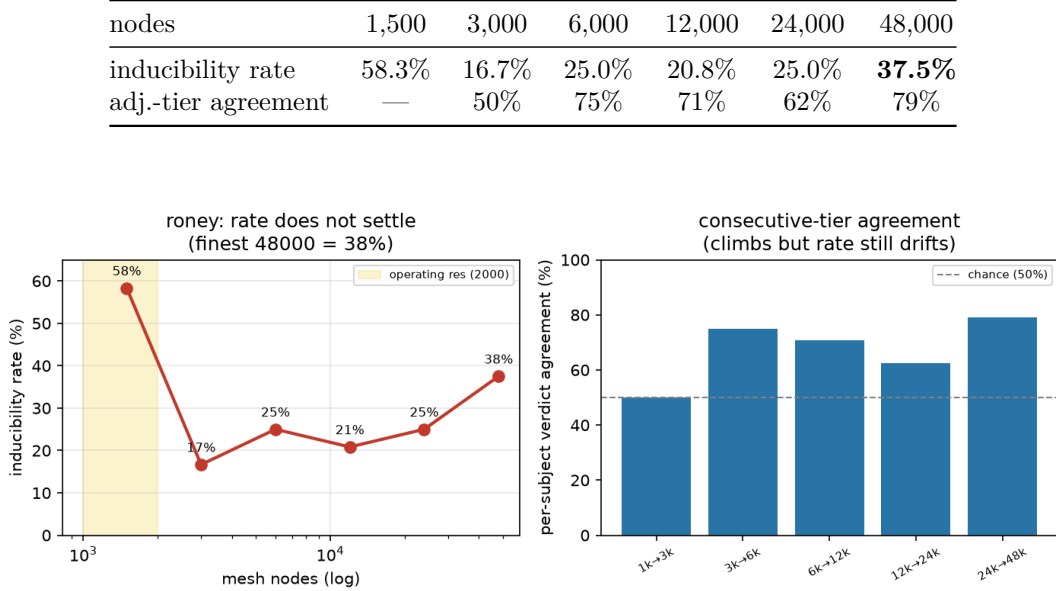


Figure 5: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement climbs (50%  $\rightarrow$  79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

**What convergence does and does not undermine.** The rate does *not* collapse to zero — the phenomenon is real; it is the *magnitude* that is resolution-sensitive up to  $\sim 50k$  nodes. Because the main results ran at 2,000 nodes, every *absolute* number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. What this does *not* touch is what we lead with: (a) the *relative* SFI-vs-competitor comparison, since both arms score the *same* labels, so this bias cancels; and (b) the label-free estimator-validity argument ( $\rho$  is computed from the graph Laplacian, no labels at all).

## 5 Discussion

**What is genuinely contributed.** (1) A pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker: a clean null at synthetic scale ( $10^5$  networks) with its mechanism (feature redundancy — even the exact  $\Delta\lambda_2$  SFI adds nothing), and an honest inconclusive/underpowered verdict on real anatomy. (2) A label-free validity criterion  $\rho$  for the single-vector estimator — it proves  $\varphi_2$  is ill-conditioned on real tissue ( $\rho \approx 2422 \gg \rho^* \approx 3$ ), a portable a-priori check for any Fiedler-based marker — with the honest scope that  $\rho$  bounds estimator accuracy, not predictive content. (3) A weak-but-real, purely connectivity-derived rank localizer that equals substrate localization and beats centrality baselines across three media. (4) A demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the  $p$ -value trap).

**Generalization beyond cardiology.** The validity screen is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability. The same precondition applies: compute  $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$ ; if  $\rho \gg 1$ , the first-order spectral marker is outside its validity radius and its in-sample correlations cannot be trusted.

**What is NOT claimed.** The mathematics is classical; no new theorem.  $\rho$  does *not* prove SFI is non-predictive — only that the single-vector estimator is ill-conditioned; the null is separately due to feature redundancy. The real-cohort result is *not* a demonstrated null — it is underpowered. The localizer is *not* precise (strict endpoint null) and does *not* beat substrate imaging. The  $\rho$  scaling “law” is a clean match to Weyl only on the real atrial 2-manifold. Labels are a monodomain simulator, not clinical outcomes and not openCARP.

## 6 Limitations

1. **The labeller over-calls inducibility at the operating mesh resolution and does not converge in the tested window** (Section 4.8): the rate drops from 58% at 1500 nodes to 17% at 3000, is non-monotone across the fine tiers, and rises to 37.5% at  $\sim 50k$  nodes; only 7/24 subjects are fully consistent across all six tiers. All *absolute* rates and AUCs are coarse-mesh-provisional; the *relative* comparison scores the same labels in both arms, so this bias cancels there.
2. Simulator, not clinical, labels: the pre-registered openCARP ground truth was substituted with monodomain Mitchell-Schaeffer, a logged deviation.
3. The UW cohort uses a PCA-surrogate universal-atrial-coordinate map and rule-based fibres, and is only  $\sim 7\%$  inducible (at/below the no-AF-control clinical floor), so its predictive test is uninformative.

4.  $\sim 30$  DeLong comparisons were run; the one positive (`fibrosis_vs_SFI`,  $p = 0.009$ ) is a pre-specified secondary, suggestive not confirmatory.
5.  $\rho^*$  and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

## 7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null), and on real anatomy the evidence is inconclusive rather than a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ( $\rho \gg 1$ ). The honest core is not a discovery but a carefully-scoped boundary — where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology would have mistaken redundant, inconclusive signal for a working biomarker. The absolute claims await the openCARP validation.

## A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel — an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals  $\mathbb{Q}$ , so `Print Assumptions` reports “*Closed under the global context*” for every theorem: **zero axioms**. A mathematically identical companion over the reals  $\mathbb{R}$  (`formal/sfi_edge_identity.v`) inherits Coq’s two standard real-number-library axioms and is kept only to show the statements hold over  $\mathbb{R}$ .

Table 5: Kernel-checked theorems ( $v = e_i - e_j$  the edge vector,  $E_{ij} = \partial L / \partial w_{ij} = vv^\top$ ). T1–T5 are axiom-free over  $\mathbb{Q}$ .

| name                                        | statement                                                                                                     | meaning                                                                                      |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| T1_rank_one_quadratic_form<br>dot_edge (T2) | $\varphi^\top (vv^\top) \varphi = (v \cdot \varphi)^2$<br>$(e_i - e_j) \cdot \varphi = \varphi_i - \varphi_j$ | quadratic form of a rank-one edge matrix<br>the edge vector selects the potential difference |
| T3_sfi_edge_sensitivity                     | $\varphi^\top E_{ij} \varphi = (\varphi_i - \varphi_j)^2$                                                     | <b>the SFI edge term</b>                                                                     |
| T4_sfi_term_nonneg                          | $0 \leq w \Rightarrow 0 \leq w(\varphi_i - \varphi_j)^2$                                                      | each SFI contribution is non-negative                                                        |
| T5_aggregate_nonneg                         | $\sum (\text{nonneg terms}) \geq 0$                                                                           | SFI is a Dirichlet energy; the Laplacian form is PSD                                         |

Combined with the classical Hellmann–Feynman theorem  $\partial \lambda_2 / \partial w_{ij} = \varphi_2^\top (\partial L / \partial w_{ij}) \varphi_2$  (Ghosh–Boyd 2006 [2], an analysis result we cite, not re-prove), T3 is exactly the closed form  $\partial \lambda_2 / \partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$ . Lean/Mathlib was attempted; its toolchain binaries are distributed only as GitHub release assets, which the compute environment’s egress policy blocks with a hard 403, so Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

## B Reproducibility

All stochastic steps are seeded; the exact configuration and commit hash accompany every reported number. Numeric verification: `python scripts/verify_sfi_identity.py` and `python`

`scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks). Convergence: `python scripts/convergence_analyze.py roney`. Figures are regenerated by `scripts/make_paper_figures.py`. Datasets: Roney LA cohort (Zenodo 5801337) and UW/Boyle meshes (Dryad 10.5061/dryad.kkwh70sg0); both are public and de-identified.

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